

A "Guess the Hidden Number" circuit with a 6-bit address bus, 4-bit data bus, 1 kHz clock, 2114 RAM, 74LS73, 74LS00, 74LS244, 74LS85, 74LS93, switches, push-buttons, and LEDs using accessible tools and parts that are in-stock.

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## Build of Materials

| Component          | Quantity | Description                                                                                                           |
|--------------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| 2114               | 1        | 1K x 4-bit static RAM IC (18-pin DIP)                                                                                 |
| LM555              | 1        | 555 Timer IC for 1 kHz clock (8-pin DIP)                                                                              |
| 74LS73             | 1        | Dual JK flip-flop for debouncing (14-pin DIP)                                                                         |
| 74LS00             | 1        | Quad 2-input NAND gate for control logic (14-pin DIP)                                                                 |
| 74LS244            | 1        | Octal buffer/line driver for output LEDs (20-pin DIP)                                                                 |
| 74LS85             | 1        | 4-bit magnitude comparator for game logic (16-pin DIP)                                                                |
| 74LS93             | 1        | 4-bit binary counter for guess tracking (14-pin DIP)                                                                  |
| SPST Switch        | 11       | 6 for address (S1-S6), 4 for data (S7-S10), 1 for read/write (S11)                                                    |
| SPST Push-Button   | 2        | LOAD button (S12), Reset button (S13, optional)                                                                       |
| LED                | 15       | 4 for input (L1-L4), 4 for output (L5-L8), 3 for feedback (L9: red, L10: yellow, L11: green), 4 for counter (L12-L15) |
| Resistor (10 kΩ)   | 22       | 11 pull-up, 11 pull-down for switches                                                                                 |
| Resistor (330 Ω)   | 15       | Current-limiting for LEDs                                                                                             |
| Resistor (10 kΩ)   | 1        | LM555 timing resistor (R1)                                                                                            |
| Resistor (68 kΩ)   | 1        | LM555 timing resistor (R2)                                                                                            |
| Capacitor (0.1 μF) | 8        | 1 for LM555 timing (C8), 7 for decoupling (C1-C7, one per IC)                                                         |
| Power Supply       | 1        | 5V DC regulated supply                                                                                                |

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## Schematic Description

Here, I describe the layout and key connections, refining the text-based schematic. The circuit is organized into functional blocks for clarity.

### 1. Power and Ground

- VCC (5V): Connects to all IC VCC pins (2114 pin 18, LM555 pin 8, 74LS73 pin 14, 74LS00 pin 14, 74LS244 pin 20, 74LS85 pin 16, 74LS93 pin 5).
- GND: Connects to all IC GND pins (2114 pin 9, LM555 pin 1, 74LS73 pin 7, 74LS00 pin 7, 74LS244 pin 10, 74LS85 pin 8, 74LS93 pin 10), LED cathodes, and switch pull-downs.
- Decoupling Capacitors (C1–C7, 0.1 μF):
  - Place one near each IC's VCC pin, connected between VCC and GND.
  - Example: C1 near 2114 (pin 18 to pin 9), C2 near LM555, etc.

### 2. Clock (LM555, 8-pin DIP)

- Purpose: 1 kHz clock for 74LS73 synchronization.
- Connections:

LM555:

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Pin 1 (GND) --- GND
Pin 2 (TRIG) --- Pin 6 (THR) & C8 [0.1μF] --- GND
Pin 3 (OUT) --- 74LS73 CLK1 (pin 1)
Pin 4 (RESET) --- VCC
Pin 5 (CTRL) --- [0.1μF] --- GND (optional)
Pin 6 (THR) --- Pin 2 & R1 [10kΩ] --- Pin 7
Pin 7 (DISCH) --- R1 [10kΩ] --- Pin 6 & R2 [68kΩ] --- VCC
Pin 8 (VCC) --- VCC & C2 [0.1μF] --- GND
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- Layout: Place LM555 top-left, with R1, R2, C8 nearby. Route OUT to 74LS73.

### 3. Control Logic (74LS73, 74LS00)

- 74LS73 (14-pin DIP, Dual JK Flip-Flop):

- Purpose: Debounces LOAD button (S12), generates pulse.
- Connections:

Pin 1 (CLK1) --- S12 (LOAD, 10kΩ pull-up to VCC, 10kΩ pull-down to GND)  
 Pin 2 (J1) --- VCC  
 Pin 3 (K1) --- GND  
 Pin 4 (CLR1) --- VCC  
 Pin 5 (Q1) --- 74LS00 pin 1  $\overline{CS}$ , pin 4  $\overline{WE}$ , 74LS93 CKA (pin 14), 74LS00 pin 9 (reset)  
 Pin 6 ( $\overline{Q1}$ ) --- Unconnected  
 Pin 7 (GND) --- GND  
 Pin 8–13 (Second FF) --- Unconnected or CLK2 to LM555 OUT (pin 3), J2 to VCC, K2 to GND, CLR2 to VCC, Q2 to 74LS00 pin 10  
 Pin 14 (VCC) --- VCC & C3 [0.1μF] --- GND

- 74LS00 (14-pin DIP, Quad NAND):

- Purpose: Generates  $\overline{CS}$ ,  $\overline{WE}$ , 74LS93 reset.
- Connections:

Gate 1 (pins 1, 2, 3):  
 Pin 1 --- 74LS73 Q1 (pin 5)  
 Pin 2 --- VCC  
 Pin 3 --- 2114  $\overline{CS}$  (pin 17)  
 Gate 2 (pins 4, 5, 6):  
 Pin 4 --- 74LS73 Q1 (pin 5)  
 Pin 5 --- S11 (read/write, 10kΩ pull-up/pull-down)  
 Pin 6 --- 2114  $\overline{WE}$  (pin 10)  
 Gate 3 (pins 8, 9, 10):  
 Pin 8 --- 74LS85 0\_A=B (pin 6)  
 Pin 9 --- 74LS73 Q1 (pin 5) or Q2 (pin 12)  
 Pin 10 --- 74LS93 R0(1) (pin 2), R0(2) (pin 3), S13  
 Gate 4 (pins 11, 12, 13):  
 Pin 11 --- S13 (reset, 10kΩ pull-down) or Unconnected  
 Pin 12 --- VCC or Unconnected  
 Pin 13 --- 74LS93 R0(1), R0(2) or Unconnected  
 Pin 14 (VCC) --- VCC & C4 [0.1μF] --- GND  
 Pin 7 (GND) --- GND

- Layout: Place 74LS73 and 74LS00 centrally, near 2114. Route Q1 to multiple ICs.

#### 4. 2114 RAM (18-pin DIP)

- Purpose: Stores 4-bit number at 6-bit address (A6–A9 grounded).
- Connections:

Pin 1 (A0) --- S1 (10kΩ pull-up/pull-down)  
 Pin 2 (A1) --- S2 (same)  
 Pin 3 (A2) --- S3  
 Pin 4 (A3) --- S4  
 Pin 5 (A4) --- S5  
 Pin 6 (A5) --- S6  
 Pin 7 (A6) --- GND  
 Pin 8 (A7) --- GND  
 Pin 15 (A8) --- GND  
 Pin 16 (A9) --- GND  
 Pin 11 (I/O1) --- S7 & 74LS85 B0 (pin 15) & 74LS244 1A1 (pin 2)  
 Pin 12 (I/O2) --- S8 & 74LS85 B1 (pin 14) & 74LS244 1A2 (pin 4)

Pin 13 (I/O3) --- S9 & 74LS85 B2 (pin 13) & 74LS244 1A3 (pin 6)  
 Pin 14 (I/O4) --- S10 & 74LS85 B3 (pin 12) & 74LS244 1A4 (pin 8)  
 Pin 10 ( $\overline{WE}$ ) --- 74LS00 pin 6  
 Pin 17 ( $\overline{CS}$ ) --- 74LS00 pin 3  
 Pin 18 (VCC) --- VCC & C1 [0.1 $\mu$ F] --- GND  
 Pin 9 (GND) --- GND

- Layout: Place centrally, with A0–A5 to switches (S1–S6), I/O1–I/O4 to switches, 74LS85, 74LS244.

#### 5. 74LS244 (20-pin DIP, Octal Buffer)

- Purpose: Buffers RAM output for L5–L8 (optional).
- Connections:

Pin 1 ( $\overline{1G}$ ) --- 74LS00 pin 3 ( $\overline{CS}$ ) & S11 (read = 1)  
 Pin 2 (1A1) --- 2114 I/O1 (pin 11)  
 Pin 4 (1A2) --- 2114 I/O2 (pin 12)  
 Pin 6 (1A3) --- 2114 I/O3 (pin 13)  
 Pin 8 (1A4) --- 2114 I/O4 (pin 14)  
 Pin 18 (1Y1) --- R12 [330 $\Omega$ ] --- L5 --- GND  
 Pin 16 (1Y2) --- R13 [330 $\Omega$ ] --- L6 --- GND  
 Pin 14 (1Y3) --- R14 [330 $\Omega$ ] --- L7 --- GND  
 Pin 12 (1Y4) --- R15 [330 $\Omega$ ] --- L8 --- GND  
 Pin 19 (2 $\overline{G}$ ) --- VCC  
 Pin 11–17 (2A1–2A4, 2Y1–2Y4) --- Unconnected  
 Pin 20 (VCC) --- VCC & C5 [0.1 $\mu$ F] --- GND  
 Pin 10 (GND) --- GND

- Layout: Place near 2114, with outputs to L5–L8 (bottom-right).

#### 6. 74LS85 (16-pin DIP, Comparator)

- Purpose: Compares guess vs. RAM data, drives L9–L11.
- Connections:

Pin 2 (A0) --- S7 & 2114 I/O1  
 Pin 3 (A1) --- S8 & 2114 I/O2  
 Pin 4 (A2) --- S9 & 2114 I/O3  
 Pin 5 (A3) --- S10 & 2114 I/O4  
 Pin 15 (B0) --- 2114 I/O1 (pin 11)  
 Pin 14 (B1) --- 2114 I/O2 (pin 12)  
 Pin 13 (B2) --- 2114 I/O3 (pin 13)  
 Pin 12 (B3) --- 2114 I/O4 (pin 14)  
 Pin 5 (O<sub>A</sub>>B) --- R16 [330 $\Omega$ ] --- L9 (Red) --- GND  
 Pin 7 (O<sub>A</sub><B) --- R17 [330 $\Omega$ ] --- L10 (Yellow) --- GND  
 Pin 6 (O<sub>A</sub>=B) --- R18 [330 $\Omega$ ] --- L11 (Green) --- GND & 74LS00 pin 8  
 Pin 10 (I<sub>A</sub>>B) --- GND  
 Pin 11 (I<sub>A</sub><B) --- GND  
 Pin 9 (I<sub>A</sub>=B) --- VCC  
 Pin 16 (VCC) --- VCC & C6 [0.1 $\mu$ F] --- GND  
 Pin 8 (GND) --- GND

- Layout: Place near 2114, with outputs to L9–L11 (right).

#### 7. 74LS93 (14-pin DIP, Counter)

- Purpose: Counts guesses, drives L12–L15.
- Connections:

Pin 14 (CKA) --- 74LS73 Q1 (pin 5)

Pin 1 (CKB) --- Pin 12 (QA)  
 Pin 12 (QA) --- R19 [330Ω] --- L12 --- GND & Pin 1 (CKB)  
 Pin 9 (QB) --- R20 [330Ω] --- L13 --- GND  
 Pin 8 (QC) --- R21 [330Ω] --- L14 --- GND  
 Pin 11 (QD) --- R22 [330Ω] --- L15 --- GND  
 Pin 2 (R0(1)) --- 74LS00 pin 10 & S13 (10kΩ pull-down)  
 Pin 3 (R0(2)) --- Same as R0(1)  
 Pin 5 (VCC) --- VCC & C7 [0.1μF] --- GND  
 Pin 10 (GND) --- GND  
 Pin 4, 6, 7, 13 (NC) --- Unconnected

- Layout: Place near 74LS85, with outputs to L12–L15 (bottom).

## 8. Switches (S1–S11)

- Connections:

S1–S6 (A0–A5):  
 Each: --- 2114 pins 1–6 & R23–R28 [10kΩ pull-up] --- VCC & R29–R34 [10kΩ pull-down] --- GND  
 S7–S10 (D0–D3):  
 S7: --- 2114 I/O1 & 74LS85 A0 & 74LS244 1A1 & R8 [330Ω] --- L1 --- GND & R35 [10kΩ pull-up] --- VCC & R36 [10kΩ pull-down] --- GND  
 S8: --- 2114 I/O2 & 74LS85 A1 & 74LS244 1A2 & R9 [330Ω] --- L2 --- GND & R37 [10kΩ] --- VCC & R38 [10kΩ] --- GND  
 S9: --- 2114 I/O3 & 74LS85 A2 & 74LS244 1A3 & R10 [330Ω] --- L3 --- GND & R39 [10kΩ] --- VCC & R40 [10kΩ] --- GND  
 S10: --- 2114 I/O4 & 74LS85 A3 & 74LS244 1A4 & R11 [330Ω] --- L4 --- GND & R41 [10kΩ] --- VCC & R42 [10kΩ] --- GND  
 S11 (Read/Write):  
 --- 74LS00 pin 5 & R43 [10kΩ pull-up] --- VCC & R44 [10kΩ pull-down] --- GND

## 9. Push-Buttons (S12, S13)

- Connections:

S12 (LOAD): --- 74LS73 CLK1 (pin 1) & R45 [10kΩ pull-up] --- VCC & R46 [10kΩ pull-down] --- GND  
 S13 (Reset): --- 74LS93 R0(1) (pin 2), R0(2) (pin 3) & R47 [10kΩ pull-down] --- GND

## 10. LEDs (L1–L15)

- Connections:

L1: S7 --- R8 [330Ω] --- Anode --- Cathode --- GND  
 L2: S8 --- R9 [330Ω] --- Anode --- Cathode --- GND  
 L3: S9 --- R10 [330Ω] --- Anode --- Cathode --- GND  
 L4: S10 --- R11 [330Ω] --- Anode --- Cathode --- GND  
 L5: 74LS244 1Y1 --- R12 [330Ω] --- Anode --- Cathode --- GND  
 L6: 74LS244 1Y2 --- R13 [330Ω] --- Anode --- Cathode --- GND  
 L7: 74LS244 1Y3 --- R14 [330Ω] --- Anode --- Cathode --- GND  
 L8: 74LS244 1Y4 --- R15 [330Ω] --- Anode --- Cathode --- GND  
 L9: 74LS85 0\_A>B --- R16 [330Ω] --- Anode (Red) --- Cathode --- GND  
 L10: 74LS85 0\_A<B --- R17 [330Ω] --- Anode (Yellow) --- Cathode --- GND  
 L11: 74LS85 0\_A=B --- R18 [330Ω] --- Anode (Green) --- Cathode --- GND  
 L12: 74LS93 QA --- R19 [330Ω] --- Anode --- Cathode --- GND  
 L13: 74LS93 QB --- R20 [330Ω] --- Anode --- Cathode --- GND  
 L14: 74LS93 QC --- R21 [330Ω] --- Anode --- Cathode --- GND  
 L15: 74LS93 QD --- R22 [330Ω] --- Anode --- Cathode --- GND

## 11. Power Supply

- Connections:

VCC: 5V --- All IC VCC pins & switch pull-ups

GND: --- All IC GND pins, LED cathodes, switch pull-downs

C1-C7 [0.1 $\mu$ F]: Between VCC and GND near each IC

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### Visual Layout for Graphical Schematic

- Top-Left: LM555 (clock) with R1, R2, C8.
  - Top-Center: 74LS73, 74LS00, S12 (LOAD), S13 (reset), S11 (read/write).
  - Center: 2114 RAM, with S1–S6 (address), S7–S10 (data), L1–L4 (input LEDs).
  - Right: 74LS85, with L9–L11 (feedback LEDs).
  - Bottom-Right: 74LS244, with L5–L8 (output LEDs).
  - Bottom: 74LS93, with L12–L15 (counter LEDs).
  - Buses:
    - 6-bit address (A0–A5): S1–S6 to 2114.
    - 4-bit data (D0–D3): S7–S10 to 2114, 74LS85, 74LS244, L1–L4.
  - Power: VCC, GND rails across top/bottom, capacitors near ICs.
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